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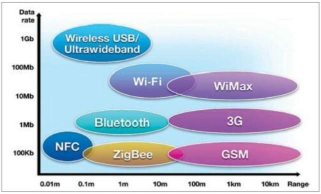

Fig. 2: Speed comparison

Improving the design. Optimising the communication receiver design leads to compromise on certain other parameters. A super-heterodyne receiver, for example, may require filtering signals between the antenna and the front-end mixer to prevent interference from unwanted sources. Here lies the problem. Once you introduce a narrow-band filter, a signal loss corresponding to receiver noise figure is also introduced. Adding a preamplifier could compensate for the filter, while lowering the noise figure.

This, on the other hand, brings along its own set of inter-modulation problems, resulting in possible degradation of mixer performance. There is still a possibility to optimise key parameters like sensitivity and dynamic range. Since adding the preamplifier results in inter-modulation problems, it creates specific design challenges.

Traditionally, high linearity and low noise in high-frequency applications was achieved by frequency applications that used gallium arsenide (GaAs) or indium phosphide (InP) gain blocks. This could become a serious problem in battery-powered or low-voltage applications because of high power dissipation. A more practical method is to improve the characteristics of the antenna and its installation to improve transmission efficiency of the overall system.

Characteristics to be considered for antenna design include frequency of transmission, input impedance, antenna gain, polarisation and radiation pattern. Provision of detachable antenna in a receiver allows you to use different types of antennae for optimal reception.

Considering the throughput. Throughput is another area to consider with wireless transmission—the higher the better. Subhas Kamble, portfolio manager communication and devices, product engineering services, Sasken, explains, "Higher throughput is addressed with newer technologies, and latency reduction increases performance."

Throughput is essentially synonymous with digital bandwidth consumption. Factors affecting throughput include analogue physical medium, available processing